Machine learning for porous materials

Advanced Methods in Digital Chemistry
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Understanding the MOF gas-prediction notebook

The uv commands we will use

macOS/Linux: Terminal. Windows: Ubuntu in WSL 2.
Use your usual browser, including your Windows browser with WSL.

```
cd digital-chemistry-course
```

Enter the extracted folder

```
uv --version
```

Check that uv is installed

```
uv sync --locked
```

Install the course environment

```
uv run --locked marimo edit --headless
```

Start the local notebook server

uv uses uv.lock by default; --locked requires it to be up to date.

--headless: start the server without automatically opening a browser.

Open the printed URL in your browser. Save; Ctrl+C stops marimo.

Our first marimo notebook

00-getting-started.py

Change one input. Predict which cells will change.

```
concentration = 2.0
prediction = slope * concentration + intercept
```

Save your own notebook copy.

Where machine learning enters

Structure — Interactions — Simulation — **Uptake**

Descriptors — Learned surrogate — **Predicted uptake**

Our labels come from completed simulations.

What are we trying to predict?

X —— **Model** —— **y**

Geometric descriptors

Simulated CO_2 uptake

298 K, 16 bar, mmol/g

A structure becomes a row of numbers

Structure

Descriptors

Material	Density	Pore volume	Pore size
A			
B			

8 inputs: 3 pore diameters, density, 2 surface areas, 2 pore volumes

Which chemical information have we discarded?

RACs add chemistry to the representation

Revised autocorrelations describe the bonding graph.

Atomic properties

$$\chi, \quad Z, \quad T, \quad S, \quad I$$

Bond distances

Combine products or differences within selected graph environments.

Original MolSim: geometry + RACs

Compact class notebook: 8 geometric features

Why we split the data

Training: 900	Validation: 300	Test: 300
Fit	Choose	Assess

Fitting uses the training answers to reduce their error.

Held-out rows test predictions on answers unused in fitting.

What counts as a genuinely new material?

Fitting adjusts the model parameters

$$\widehat{y} = w_0 + w_1 x$$

Predictions



Training loss



Compare with training targets, then update the parameters.

Several descriptors give a linear model

$$\widehat{y_i} = w_0 + w_1 x_{i1} + \cdots + w_p x_{ip}$$

Here: 8 geometric descriptors

The model fits the weights and intercept using training rows.

A baseline and one error calculation

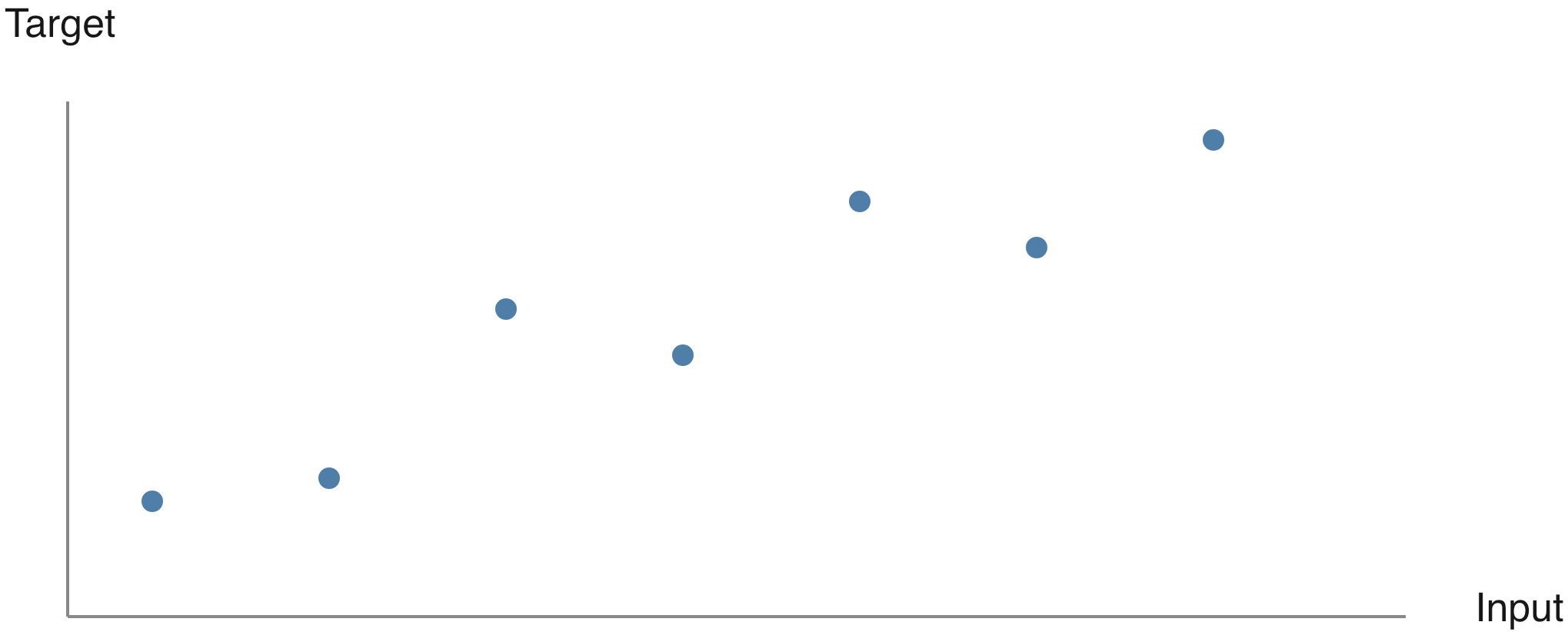
Baseline: median training uptake

$$\mathrm{MAE} = \frac{1}{n}\sum_{i=1}^{n} |\widehat{y}_i - y_i|$$

$$\text{Residuals: } -2,\ 0,\ +1\ \mathrm{mmol\,g^{-1}}$$

MAE =

Training error can keep falling



Which curve will predict the next observation?

Bias and variance explain different failures

Bias

The average fit misses
the relationship.



Variance

The fit changes when
the training sample changes.



Fresh training sets → refit the same learning procedure

Regularisation constrains the fit

Ridge example

$$\underbrace{\sum_{i \in \text{train}} (y_i - \widehat{y}_i)^2}_{\text{training loss}} + \alpha \underbrace{\sum_{j=1}^{p} w_j^2}_{\text{ridge penalty}}$$

Weak regularisation

Strong regularisation

Validation helps choose the regularisation strength.

Nonlinear features can bend a linear model

Features: $x, \quad x^2$

$$\widehat{y} = w_0 + w_1 x + w_2 x^2$$



Linear in the fitted coefficients

A kernel gives us a way to work with a richer feature space.

The kernel trick avoids explicit feature expansion

$$\mathbf{x} = (x_1, x_2) \qquad \phi(\mathbf{x}) = (x_1^2, \sqrt{2}\, x_1 x_2, x_2^2)$$

Two input coordinates become three feature coordinates.

$$k(\mathbf{x}, \mathbf{z}) = \phi(\mathbf{x})^\top \phi(\mathbf{z}) = (\mathbf{x}^\top \mathbf{z})^2$$

Evaluate the inner product without constructing the features.

Two calculations give the same kernel value

$$\mathbf{x} = (1, 2), \qquad \mathbf{z} = (3, 4)$$

$$\phi(\mathbf{x}) = (1, 2\sqrt{2}, 4) \qquad \phi(\mathbf{z}) = (9, 12\sqrt{2}, 16)$$

$$\phi(\mathbf{x})^\top \phi(\mathbf{z}) = _____$$

$$(\mathbf{x}^\top \mathbf{z})^2 = _____$$

The Gaussian kernel compares descriptor vectors

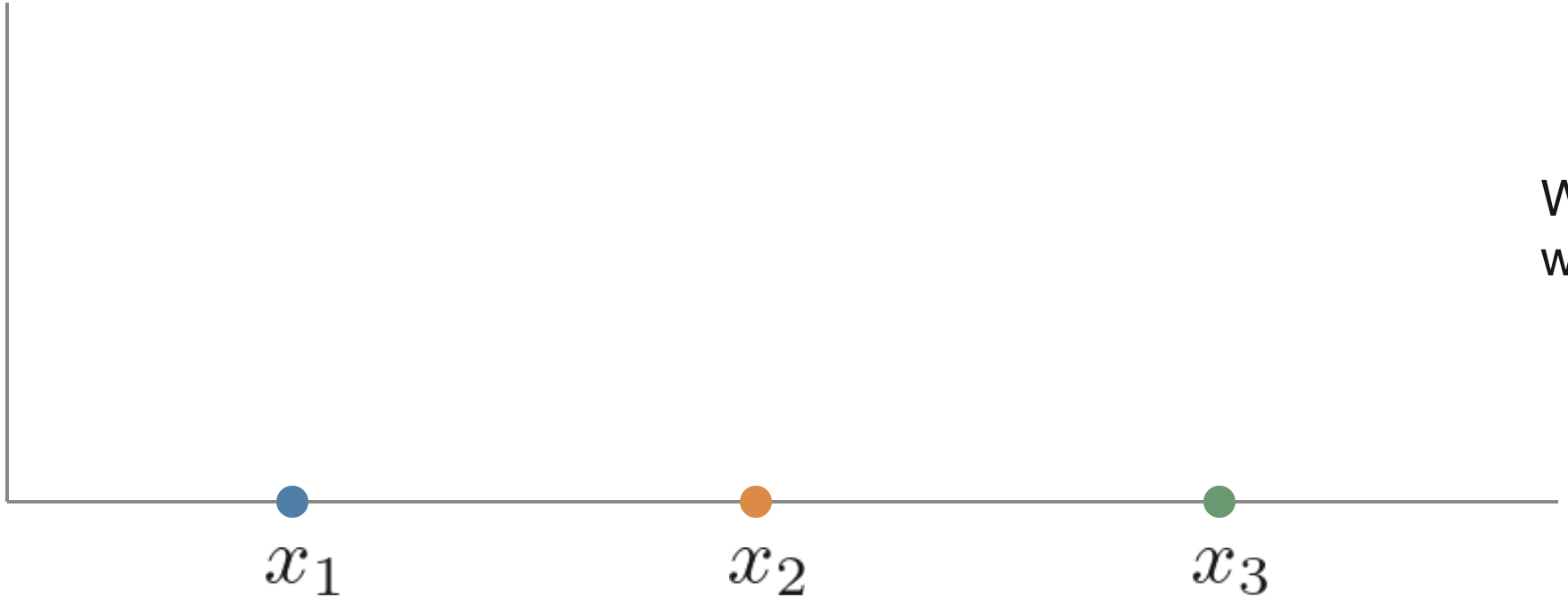
$$k(\mathbf{x}, \mathbf{z}) = \exp\left(-\gamma \|\mathbf{x} - \mathbf{z}\|^2\right)$$

x and z each contain the same 8 descriptors.

Larger gamma gives a narrower neighbourhood.

A kernel describes similarity

One contribution centred on each training input



Width changes
what “near” means.

A kernel prediction adds fitted contributions

Training point	Fitted coefficient	Similarity to query	Contribution
1	2	0.8	
2	−1	0.2	
3	3	0.05	

Prediction =

Illustrative numbers, arbitrary target units

Units change numerical distances

Difference in density	**0.1 g/cm^3**
Difference in surface area	**100 m^2/g**

Squared distance =

Standardisation uses training statistics

$$z_j = \frac{x_j - \mu_j}{\sigma_j}$$

One mean and standard deviation for each feature

Estimate both statistics from the training rows.

Validation and test use the same transformation.

A pipeline keeps scaling and fitting together

Training rows ——— Fit scaler ——— Fit model

Validation rows —— Transform ——— Predict and score

make_pipeline(StandardScaler(), KernelRidge(...))

Validation and test use the scaler fitted on training rows.

Validation chooses the settings

Training data

Fit model parameters

Validation data

Compare candidate settings

The lowest validation error can partly reflect luck.

Use validation to choose; use the untouched test to assess.

The notebook compares four candidates

Model	Scaling	Prediction	Validation MAE
Training median	None	Constant	
Ridge	Training statistics	Linear	
KRR	None	Gaussian kernel	
KRR	Training statistics	Gaussian kernel	

Ridge and KRR: $\alpha = 1$. Both KRR models: $\gamma = 0.1$.

Which comparison isolates the effect of scaling?

Cross-validation repeats the comparison

Fit 1	Validate	Train	Train	Train	Train
Fit 2	Train	Validate	Train	Train	Train
Fit 3	Train	Train	Validate	Train	Train
Fit 4	Train	Train	Train	Validate	Train
Fit 5	Train	Train	Train	Train	Validate

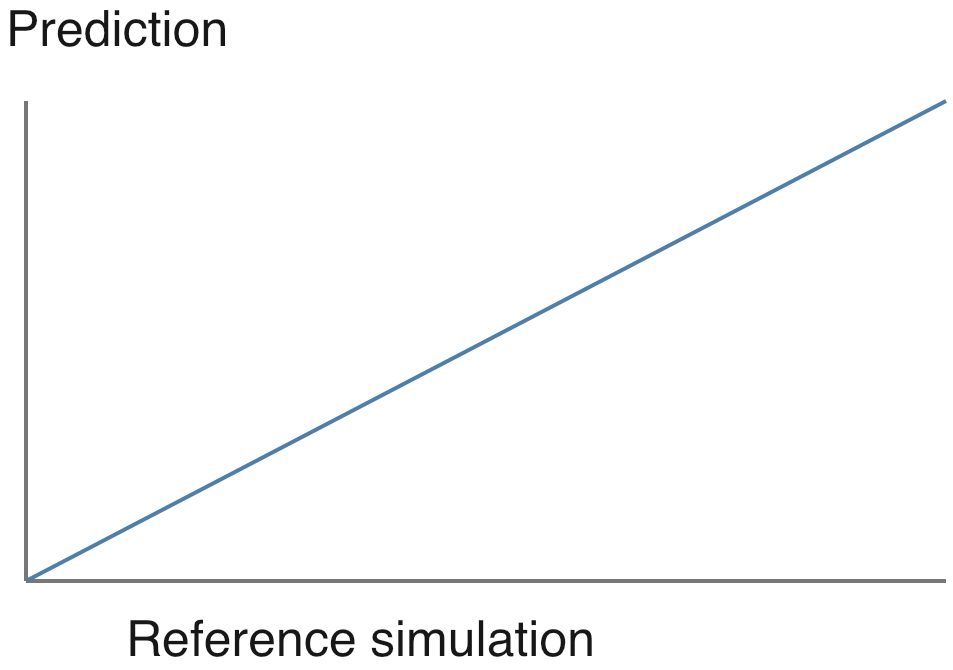
Held-out scores

Mean =

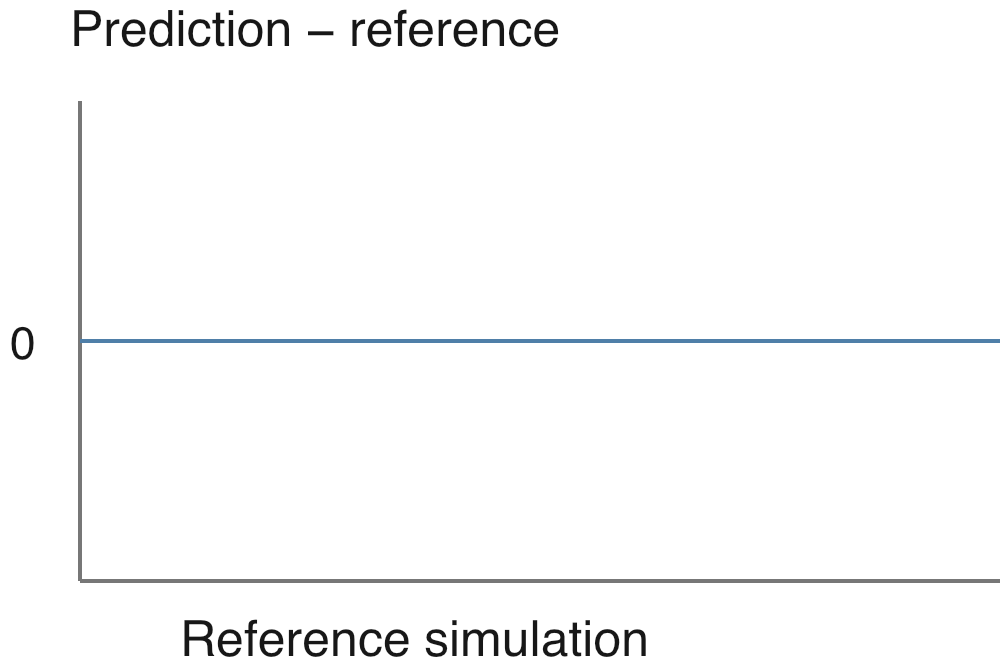
CV extends the notebook’s single validation split.

Reading the notebook’s prediction plots

Parity plot



Residual plot



Validation rows. All values in mmol/g.

The final test answers a narrower question

Record the model choice, then reveal the test result.

Which materials does this result describe?

300 held-out rows. MAE in mmol/g.

A test error is an empirical estimate

What we want to know

$$R(\widehat{f}) = \mathbb{E}_{(X,Y)\sim P}\left[\left|\widehat{f}(X) - Y\right|\right]$$

Average error over future MOFs from the intended population.

What we can calculate

$$\widehat{R}_{\text{test}}(\widehat{f}) = \frac{1}{n_{\text{test}}} \sum_{i \in \text{test}} \left|\widehat{f}(\mathbf{x}_i) - y_i\right|$$

Average error on 300 reserved rows.

Keep the model fixed. A new test sample can give a different estimate.

Hands-on with the ML MolSim material

03-mof-uptake.py

Baseline

Unscaled and scaled KRR

Record the validation comparison before the test.

Supported practical: 2 December, 12:00–16:00

A project starts with a chemical question

What would you like to find out?

Which data could you use?

What would be a useful first comparison?

Add an idea to the document in the course email.